SMIT BANGARE

AI / ML Engineer
Mumbai, India | +91 93723 56521 | smitbangare@gmail.com
LinkedIn: linkedin.com/in/smit-bangare | GitHub: github.com/SmitBangare

SUMMARY

AI & ML engineer with hands-on experience across agentic AI, generative AI (RAG, LLM agents), and computer vision, spanning a full-time applied AI role, a physics-based research internship at TIFR, and multiple deployed ML systems. Strong foundation in deep learning, multi-agent orchestration, and end-to-end ML pipeline development.

EDUCATION

B.Tech Electronics and Computer Engineering | 2022 – 2026
K. J. Somaiya College of Engineering, Mumbai CGPA: 8.43 / 10

Higher Secondary Certificate (Class XII), Science | 2021
Khar Education Society, Mumbai 92.6%

EXPERIENCE

AI / ML Engineer — Tech Shipping OPC Pvt. Ltd., Mumbai
June 2025 – August 2026 · Full-time

- Built an **AI-powered Maritime Procurement & Port Intelligence platform** digitising ship-chandling operations across Mumbai, JNPT and 8+ Maharashtra/Gujarat ports.
- Developed an **AI Operations Copilot** using RAG architecture and LLM agents over supplier, procurement, inventory and vessel records for natural-language operational queries.
- Delivered **predictive ML models** for vessel ETA, demand forecasting, supplier scoring and delivery-risk prediction. Automated the **end-to-end procurement pipeline** with AI document extraction from emails, PDFs and quotations.

Research Intern — Tata Institute of Fundamental Research (TIFR), Mumbai
January 2026 – May 2026 · Part-time

- Developed a **physics-based regression model** for DC beamline magnet calibration at the TIFR Pelletron Linac Accelerator, derived from the magnetic rigidity relation.
- Applied **OLS regression** on 853 logbook entries with a three-stage filtering pipeline and LOOCV, achieving **sub-1% prediction error** for BM1, BM2 and 14° magnet setpoints.

Machine Learning Intern — Unified Mentor Pvt. Ltd., India
December 2024 – January 2025

- Built and evaluated predictive ML models across multiple real-world datasets; delivered data visualisation dashboards and documented end-to-end ML workflows.

Research and Strategy Intern — Skylark Realty Pvt. Ltd., India
June 2024 – August 2024

- Contributed to a **smart HVAC optimisation prototype** using PatchTST for energy load forecasting and dynamic zoning from real-time occupancy data; delivered a scalable ML framework for infrastructure cost optimisation.

KEY PROJECTS

Multi-Agent Medical Imaging Diagnostic System | 2025

- Engineered a production-grade **multi-agent AI orchestration framework** using LangGraph for medical imaging diagnosis — specialised agents for pneumonia detection, sepsis risk assessment, and multimodal clinical data fusion. Built an **Agentic RAG** layer with Qdrant hybrid retrieval and HuggingFace reranking. Deployed via FastAPI with Docker and GitHub Actions CI/CD; capstone prepared for conference publication.

Aeronext AI Platform — Smart HVAC Forecasting for Airports | 2025

- Developed an **AI-driven airport platform** integrating HVAC load forecasting using PatchTST transformers with passenger-centric revenue optimisation via reinforcement learning. Delivered real-time Streamlit dashboards. Extended as research under Dr. Om Prakash Goswami; prepared for journal submission.

Context-Aware Edge Detection using Hybrid CNN–Transformer Architectures | 2025

- Research under Dr. Om Prakash Goswami replacing VGG-16 with ResNet-50/EfficientNet and incorporating a **transformer-based refinement block** for long-range spatial context. Optimised Dice/Focal Loss and evaluated on BSD500, outperforming baseline HED models in boundary precision.

VigilPay — Intelligent Transaction Security Solution for India | 2024

- Built a **smart fraud prevention platform** with real-time transaction monitoring, AI-driven risk assessment, and adaptive fraud detection for Indian spending patterns and RBI/UPI compliance. Developed with Next.js, MongoDB, JWT authentication. **Finalist, Smart India Hackathon 2024.**

AI For an Eye — Smart Assistive Glasses for the Visually Impaired | 2024

- Engineered a real-time **computer vision pipeline** for smart assistive glasses with multi-class object and human detection and end-to-end text-to-speech feedback using OpenCV and PyTorch. Validated across real-world environments as part of a cross-functional team.

TECHNICAL SKILLS

Languages: Python, C, Java, SQL, JavaScript
Libraries & Frameworks: NumPy, Pandas, TensorFlow, PyTorch, Scikit-learn, OpenCV, Keras, LangChain, LangGraph, LlamaIndex, Hugging Face, FastAPI, Qdrant, Pydantic
Tools: Google Colab, MySQL, MATLAB, Docker, Git, GitHub Actions
AI & Data Science: Deep Learning, Machine Learning, Computer Vision, NLP, Generative AI, Multi-Agent Systems, Agentic RAG, Vector Databases